

VayuVeer Cloud GCS — Raspberry Pi 5

Installation Guide

Companion computer setup for internet control of the AiRotor AX4 · Relay: <https://gcs.vayuveer.in> · Airotor Labs, Ahmedabad

What you need: Raspberry Pi 5 with Raspberry Pi OS Bookworm (64-bit) on a microSD card · a 5 V / 5 A BEC or power module · internet for the Pi (Wi-Fi, or a 4G/5G USB dongle for flight) · 3 jumper wires to the flight controller (or a USB cable) · the **access token** and an **aircraft ID** from your administrator.

1 Wire the Pi to the flight controller

Connect the flight controller's **TELEM2** port to the Pi's 40-pin header. Both are 3.3 V logic — no level shifter.

FC TELEM2	Raspberry Pi 5 header
TX	GPIO15 RXD — physical pin 10
RX	GPIO14 TXD — physical pin 8
GND	GND — physical pin 6

Alternative: plug the FC's USB port into the Pi.
The installer detects it as `/dev/ttyACM0`.

Pi 5 header (top-left corner of the board, SD card side)

3V3	(1)	(2)	5V	
GPIO2	(3)	(4)	5V	
GPIO3	(5)	(6)	GND	← FC GND
GPIO4	(7)	(8)	TXD	← FC RX
GND	(9)	(10)	RXD	← FC TX
...				

Power the Pi from its own 5 V / 5 A supply. Do not power it from the flight controller's TELEM port.

2 Set the flight controller parameters (ArduPilot Copter)

Parameter	Value	Purpose
SERIAL2_PROTOCOL	2	MAVLink 2 on TELEM2
SERIAL2_BAUD	921	921600 baud
FS_GCS_ENABLE	1	Return-to-launch if the Pi stops sending heartbeats
GUID_TIMEOUT	3 (default)	Stops joystick velocity commands if they are not refreshed

Set them in Mission Planner or QGroundControl, write parameters, then reboot the flight controller.

3 Install the software (double-click)

1. Copy `VayuVeer-RPi-Installer.sh` onto the Pi (USB stick, or download it from the administrator).
2. Double-click the file on the Pi desktop and choose **Execute in Terminal**.

3. Answer the questions: **access token** (skipped if your copy is pre-filled), **aircraft ID** (for example `vayuveer-01`), and the **camera** type (Pi camera module, USB camera, or none).
4. Wait for “Done”, then answer **Y** to reboot.

The installer downloads the software, installs its dependencies, enables the serial port to the flight controller, writes the configuration, and installs a service that starts automatically on every boot.

Manual alternative, from a terminal on the Pi:

```
git clone https://github.com/airotorac/vayuveer-cloud-gcs.git ~/vayuveer-cloud-gcs
&& cd ~/vayuveer-cloud-gcs && ./drone/install_rpi.sh
then edit drone/config.yaml (server_url, token, drone_id, drone.mode: mavlink, camera source) and
reboot.
```

4 Check that it is running

After the reboot, open a terminal on the Pi and run:

```
journalctl -fu vayuveer-agent
```

You should see `heartbeat` from `system 1` (flight controller link) and `relay connected` (cloud link). Press Ctrl+C to stop watching. The service keeps running and restarts by itself after any crash or reboot.

5 First flight test

Step	What to do	Expect
Bench, props off	Sign in at <code>https://gcs.vayuveer.in</code> , pick the aircraft in the header dropdown. Arm, then Disarm.	Relay, Aircraft and FC pills green; telemetry matches the FC; video flowing; motors respond to Arm/Disarm.
Open area, props on	Arm → Take off to 5 m → Hold → Land. Then try the joysticks gently in Hold.	Stable hover; joystick moves the aircraft at walking pace; releasing the stick stops it.
Failsafe, low altitude	Unplug the internet dongle while hovering.	After 10 s the aircraft returns home and lands by itself.

6 Troubleshooting

Symptom	Fix
<code>heartbeat lost</code> / FC pill not green	Check TX and RX are crossed, the FC parameters above, and that <code>/dev/serial0</code> exists.
camera <code>'picamera2'</code> failed	Test the camera with <code>rpicas-hello</code> . For a USB camera re-run the installer and choose camera type 2. The aircraft still flies with a simulated feed.
<code>relay link down</code>	The Pi has no internet, or the access token is wrong (the relay closes the link with code 4401).

Video stutters on 4G	Lower fps or width under camera: in ~/vayuveer-cloud-gcs/drone/config.yaml, then sudo systemctl restart vayuveer-agent .
Update the software	cd ~/vayuveer-cloud-gcs && git pull && sudo systemctl restart vayuveer-agent

VayuVeer Cloud GCS · github.com/airotorac/vayuveer-cloud-gcs · Keep the access token confidential: anyone holding it can connect an aircraft to the relay.